

Program : Diploma in Artificial Intelligence	
Course Code : 3382	Course Title: Introduction to Artificial Intelligence
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

Students will be able to summarize the fundamental concepts of artificial intelligence and solve various problems using AI techniques.

Course Prerequisites:

Topic	Course name	Semester
Basic knowledge in mathematics	Mathematics I	1
	Mathematics II	2
Basic concepts of programming	Introduction to IT systems lab	1
	Problem Solving and Programming	2

Course Outcomes

On completion of the course, the students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of Artificial Intelligence and the significance of Intelligent Agents in AI.	15	Understanding
CO2	Apply Mathematical logic for knowledge representation of AI	15	Applying
CO3	Practice various search techniques in AI	16	Applying
CO4	Describe learning and expert systems.	14	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of Artificial Intelligence and the significance of Intelligent Agents in AI.		
M1.01	Define AI, foundation and history of AI.	2	Understanding
M1.02	Discuss applications of AI.	3	Understanding
M1.03	Explain the Turing test.	1	Understanding
M1.04	Explain AI problems	3	Understanding
M1.05	Explain agents and environments.	2	Understanding
M1.06	Identify structure of agents	2	Understanding
M1.07	Recognize types of agents.	2	Understanding
Contents: Introduction to AI- AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents- Agents and Environments, Structure of Agent - types of agents.			
CO2	Apply Mathematical logic for knowledge representation of AI		
M2.01	Summarize Ontologies, foundations of knowledge representation and reasoning	3	Understanding

M2.02	Describe the representation and reasoning about objects, relations, events, actions, time and space	2	Understanding
M2.03	Demonstrate Propositional Logic	2	Applying
M2.04	Demonstrate First Order Logic	2	Applying
M2.05	Explain Soundness and Completeness,	2	Understanding
M2.06	Explain Forward and Backward chaining	2	Understanding
M2.07	Familiarize with classical planning	2	Understanding
	Series Test – I	1	

Contents:

Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space.

Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining.

Classical Planning - Algorithms for planning state space search, Planning Graphs.

CO3	Practice various search techniques in AI		
M3.01	Summarise types of search algorithms	2	Understanding
M3.02	Implement Breadth First Search and Depth First Search	6	Applying
M3.03	Implement Best first search, Hill Climbing, A* & Beam Search	6	Applying

Contents:

Search Algorithm Terminologies, Types of search algorithms

Uninformed Search: Breadth First Search, Depth First Search

Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm
Beam Search

CO4	Familiarize with constraint satisfaction problems in AI		
M4.01	Solve optimal paths - Branch & Bound, Divide & Conquer approaches, Beam Stack Search	3	Applying

M4.02	Describe the Problem Decomposition techniques - Goal Trees, AO*, Rule Based Systems	4	Applying
M4.03	Compare and summarize game playing algorithms - Minimax, AlphaBeta	4	Applying
M4.04	Familiarize Constraint Satisfaction Problem	3	Understanding
	Series Test – II	1	

Contents:

Finding optimal paths -Branch & bound, Divide & Conquer approaches

Problem Decomposition: Goal Trees, AO*, Rule Based System.

Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm.

Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems.

Text / Reference:

T/R	Book Title / Author
T1	Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill
T1	Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition. Prentice Hall.
R1	Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India)
R2	Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann
R3	Donald A. Waterman, 'A Guide to Expert Systems', Pearson Education

Online Resources:

Sl. No.	Website Link
1	https://www.tutorialspoint.com/virtualization2.0/index.htm
2	https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
3	https://onlinecourses.swayam2.ac.in/arp19_ap79/preview